

MongoDB Atlas Free Tier Access Guide

Purpose

Give students a fast, repeatable process to spin up a no-cost MongoDB Atlas cluster for coursework or prototypes. These steps assume no prior Atlas experience and can be completed in ~15 minutes.

1. Create/Sign In to Atlas

1. Browse to <https://www.mongodb.com/cloud/atlas/register>.
2. Sign up with a school email (recommended) or single sign-on (Google/GitHub).
3. When prompted for a plan, choose **Shared** (the free M0 tier). This provides 512 MB storage and is sufficient for labs/projects.

2. Set Up Organization & Project

1. Atlas automatically creates a default organization. Rename it to match the course (e.g., CST4714 Spring 25).
2. Within that org, create a **Project** named after your team or assignment. Projects isolate clusters and access control.
3. Invite teammates via **Access Manager** → **Project Access** → **Add New User**. Grant the Project Data Access Read/Write or Project Owner role as needed.

3. Build the Free Cluster

1. In the project dashboard, click **Build a Database** → **Shared**.
2. Leave the **M0 Sandbox** size selected. Pick the nearest AWS/Azure/GCP region to minimize latency.
3. Choose the latest MongoDB version unless the assignment specifies otherwise.
4. Name the cluster clearly (e.g., team-rocket-m0). Provisioning takes 2–3 minutes.

4. Configure Security

1. **Database User:** Go to **Database Access** → **Add New Database User**.
 - Use a unique username (no spaces) and generate a strong password.
 - Assign the `Atlas admin` or `readWriteAnyDatabase` role for coursework. Store the credentials in your team password vault.
2. **Network Access:** Navigate to **Network Access** → **Add IP Address**.
 - For quick testing, click **Allow Access From Anywhere (0.0.0.0/0)**, but remove it once finished.
 - Prefer adding your current IP or a VPN block to limit exposure.
3. Enable **Built-in Encryption + TLS** (default). Do not disable TLS unless using trusted private networks.

5. Connect to the Cluster

1. From the cluster card, click **Connect** → **Drivers**.
2. Select your driver (Node, Python, etc.) to copy the formatted connection string.
3. Replace <password> with the database user password and <dbname> with your target database name (e.g., admin).
4. Test locally:

```
mongosh "mongodb+srv://<username>:<password>@team-rocket-m0.mongodb.net/?retryWrites=true&w=majority"
```

5. For GUI access, download MongoDB Compass and paste the same URI under **New Connection**.

6. Seed Sample Data (Optional)

- Atlas → **Browse Collections** → **Add My Own Data** to import JSON/CSV.
- Or, run mongorestore/mongoimport from your machine by appending &tls=true&retryWrites=true&w=majority to the URI.

7. Cost Controls & Housekeeping

- M0 is always free, but ensure you stay on the **Shared** tier (no credit card needed).
- Pause/delete clusters when the semester ends: ... → **Terminate**.
- Regularly rotate database user passwords and prune unused users/IP entries.

Troubleshooting Tips

Issue	Fix
IP address not whitelisted	Re-add your current IP under Network Access or use the 0.0.0.0/0 toggle temporarily.
Authentication failed	Confirm username/password, ensure special characters are URL-encoded in the connection string.
Slow queries	Verify you are on an M0 cluster (512 MB). Consider cleaning up data or upgrading temporarily to M2/M5 (credit card required).
Driver TLS errors	Update your driver/client to the latest version so it supports SRV/TLS defaults.

Share with Your Team

Once working, drop the final URI (without password) plus setup notes into your team README. Store credentials in a secure secrets manager, not in plain text.